



Installation and operation manual for transformer

May 2026

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1. Brief Introduction

1.1 Transport

- Select appropriate transportation tools according to the overall dimensions and weight of the products.
- The products shall be placed horizontally during transportation. The inclination of the long axis of the transformer shall be $\leq 10^\circ$. Prevent the product from collision, deformation or scratch.
- Highway transportation requirements:
 - lane width $\geq 3M$;
 - Maximum gradient 1:10;
 - The shortest distance of stopping sight is 25m;
 - The minimum traffic height is required to be 4.5m;
 - Solid subgrade and smooth pavement;
 - The transportation vehicle shall keep driving at a constant speed, and the speed control:
 - Expressway $\leq 70\text{km/h}$, curve $\leq 50\text{km/h}$;
 - Ordinary highway $\leq 50\text{km/h}$, curve $\leq 30\text{km/h}$;
 - Site road $\leq 30\text{km/h}$, curve $\leq 5\text{km/h}$;
 - Vibration in three-dimensional direction $\leq 3G$;
 - The pressure shall be released once every 1000m of altitude change during transportation.
- Railway transport requirements:
 - 1) Transformers installed on railway flat cars or transport vehicles, including packaging and cable fixing devices, shall be within the specified outer limits of railway transportation.
 - 2) The loading position and cable fixation of transformer on railway flat car or transport vehicle must ensure that all wheels bear uniform load. During transportation, the transformer shall not move or overturn under the maximum allowable load.
- Water transport requirements: can meet the full hull navigation requirements.
- Air transport requirements: containers can be helicopter lifting transport, helicopter lifting weight should be more than 35t.
- Equipped with transformer collision recorder: according to the transportation requirements, simple collision recorder.

1.2 Check Delivery

- Check whether the delivered parts are complete, whether the screws are loose and whether there is any obvious external damage. If there are any missing items, loose screws or any damage, please contact your dealer. Please check the packing list for the delivered parts distributed with the box.

1.3 Storage

- This product is designed according to the requirements of long-term operation in outdoor environment, so it can meet the requirements of long-term storage in outdoor environment without shelter, but indoor storage is recommended for long-term storage;
- When receiving the transformer, necessary inspection shall be carried out:
 - The transformer shall be free from obvious collision deformation and paint damage through visual inspection;
 - The oil level display of transformer shall be normal;
 - Check whether the transformer has oil leakage.
- Necessary isolation and protection shall be taken to prevent the transformer from accidental collision damage during storage;
- The transformer shall be stored in an acidic and humid environment, and the corrosion caused by high temperature and high ventilation shall be avoided;
- Patrol inspection shall be maintained frequently during storage, and the actual patrol frequency shall be determined by the storage unit according to the complexity and stability of the storage environment. When the warehouse is locked for a long time, it can be inspected once every three months.
- Storage temperature: $-40 \sim +70^\circ\text{C}$; Storage humidity: 5%-95%RH

2.Install

2.1 Installation Environment Requirements

- The altitude shall not exceed 3,000 m.
- The installation ambient temperature range is:-45℃~+45℃.
- During the installation, the outdoor wind speed shall not exceed 30 m/s.
- Relative humidity: the daily average is not greater than 95%, and the monthly average is not greater than 90%.
- Places without fire, explosion hazard, serious pollution, chemical corrosion and violent vibration.

2.2 Electrical Installation

2.2.1 Prepare the cable

Precautions when selecting cables:

- The cable needs to have sufficient flow carrying capacity, and the cable overload work is strictly prohibited.
- Cable on the same side shall be of the same specification and type.

2.2.2 Connect the protected ground line

- Grounding should comply with local electrical safety regulations of the project.
- Before carrying out any operation on the equipment, it must be ensured that it has been grounded; When dismantling the equipment, finally remove the grounding wire.
- It is strictly prohibited to install fuses, switches, and other equipment on the ground wire.

2.2.3 Operation transformer

- Adjust the no-load tap changer according to user needs.
- Remove the safety pressure plate of the pressure relief valve.
- Install the moisture absorber (if present)
- Check if there is gas inside the gas relay, and if so, release the corresponding gas (if any)
- If the oil level indication of the transformer is higher or lower than the warning oil level line, the transformer should be drained or replenished with oil.

2.2.4 Connecting the AC input line

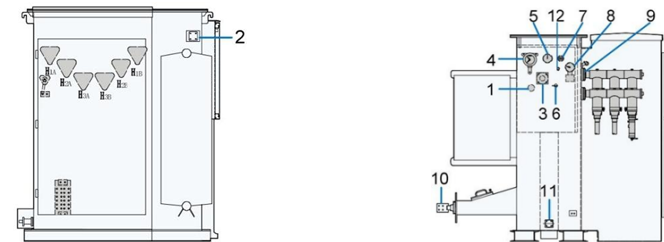
Grounding requirements

- The cables are arranged neatly and do not damage the cable skin.
- The cable joint or bus connection made on the site must be standard, firm, reliable and beautiful.
- The connection of power lines, ground lines and various signal lines shall be strong and reliable; the cables shall not bend excessively to avoid damage to the core, and conform to the system routing plan.

3.Common operating parts

3.1 Common Parts

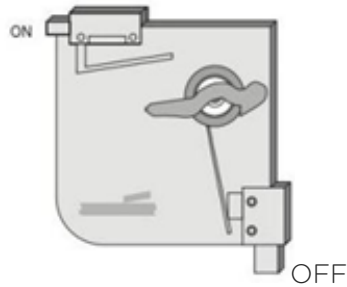
(Show the position of the whole component on the transformer overall.)
Please refer to the figure below.



No.	Name
1	Load tap changer
2	Injection valve
3	Pressure release valve 1
4	Oil thermometer
5	Pressure gauge
6	Two-stage load switch
7	Pressure release valve 2
8	Oil level indicator
9	High voltage bushing
10	Low voltage bushing
11	Oil drain valve
12	Nitrogen injection valve

3.1.1 Load switch (If have)

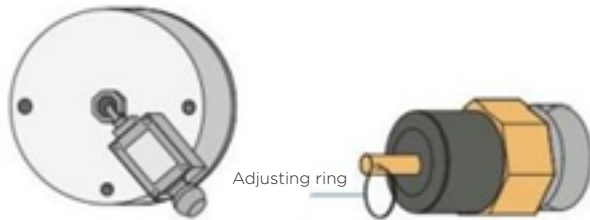
Illustration



Function: The American box transformer adopts the oil-immersed two-position load switch, which can switch under the no-load current, but cannot be used to open the fault current. No load switch, only with no load.

3.1.2 Pressure release valve

Illustration



Function: The transformer is equipped with pressure release valve. When the internal pressure of the transformer is caused by the transformer overload or fault, the pressure release valve moves to release the pressure. When the pressure decreases to the closing pressure value of the valve, the valve is closed reliably to effectively prevent external air, water and other impurities from entering the tank.

3.1.3 No load split switch

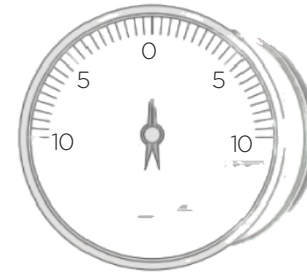
Illustration



Function: The transformer high voltage side is equipped with the tap voltage position indicator to observe the high voltage tap voltage gear at any time. Under no-excitation conditions, the number of primary coil turns of the transformer is changed according to the line side voltage, that is, the primary gear voltage of the transformer is consistent with the line side, so as to achieve the purpose of stabilizing the secondary voltage. Unexcitation tap switch is strictly prohibited from live operation.

3.1.4 Pressure gauge (if have)

Illustration



Function: Measure the pressure in the oil tank for the monitoring function.

3.1.5 Oil surface thermometer

Illustration



Function: The transformer is equipped with an oil surface thermometer, which can observe the oil surface temperature at any time. At the same time, with optional signal alarm / trip contact, and PT100 or 4-20mA or RS485 remote transmission signal, control the corresponding cooling system, to achieve the purpose of controlling the transformer temperature rise.

3.1.6 Oil level indicator

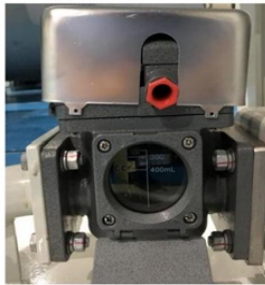
Illustration



Function: The transformer is equipped with an oil level gauge. Indicates the height of the oil level inside the transformer, along with optional electrical signal contacts.

3.1.7 Gas relay (If have)

Illustration



Function: The transformer is equipped with a gas relay. During the failure inside the transformer, the gas decomposed due to insulation damage forces the light gas of the gas relay/heavy gas contact, and sends a slight alarm signal/heavy gas trip to the control room. The operator shall immediately take measures to eliminate the fault, or trip the circuit breaker automatically to prevent the accident from further expanding.

3.1.8 Drain Oil+Sampling Valve (if have)

Illustration



Function: an integrated oil drain/sampling valve is installed at the bottom of the oil tank, which can be used for oil drain or oil sampling.

3.1.9 Winding Thermometer (if have)

Illustration



Function: The transformer is equipped with a winding thermometer, which can observe the winding temperature at any time. At the same time, it has an optional alarm/trip signal contact, and a PT100 or 4-20mA or RS485 remote transmission signal, to control the corresponding cooling system, to achieve the purpose of controlling the transformer temperature rise.

3.2 Common Operation

3.2.1 How to adjust the unexcitation tap switch

The unexcitation tap switch can adjust the output voltage of the transformer. When operating the unexcitation tap switch, please ensure that the transformer is in the unexcitation state, that is, the high and low voltage sides of the transformer are not charged.

When the low voltage voltage remains unchanged, the output voltage at different gears is:

- Gear 1: normal voltage*1.05
- Gear 2: normal voltage*1.025
- Gear 3: normal voltage
- Gear 4: normal voltage*0.975
- Gear 5: normal voltage*0.95

Step: Illustration



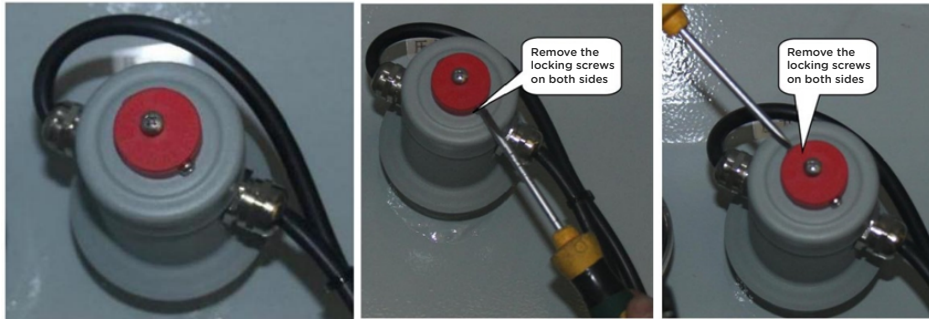
Unscrew the protective cover of the tap changer on the transformer and place the locating pin in the neutral position.



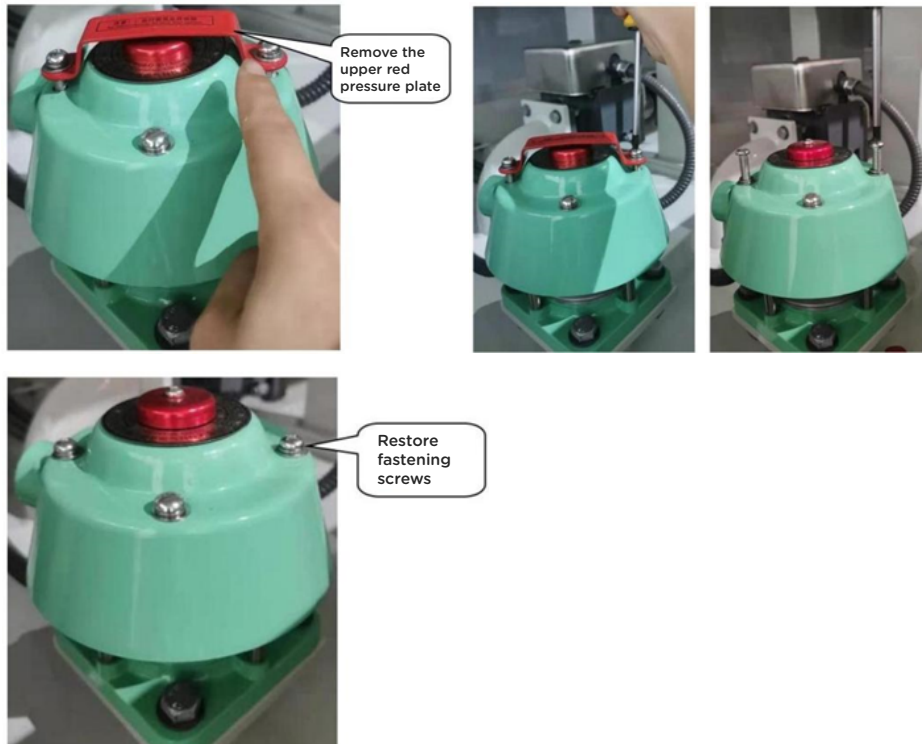
When adjusting the gear position, adjust the split switch to the corresponding position according to the output voltage.

3.2.2 Remove the pressure plate of the pressure release valve

1) YSF6 pressure valve: The locking screws on both sides of the pressure release valve must be removed according to the following drawing before operation.



2) Type YSF4 pressure valve: you must remove the upper red pressure plate of the pressure release valve according to the drawing below, and tighten the screws again before operation.

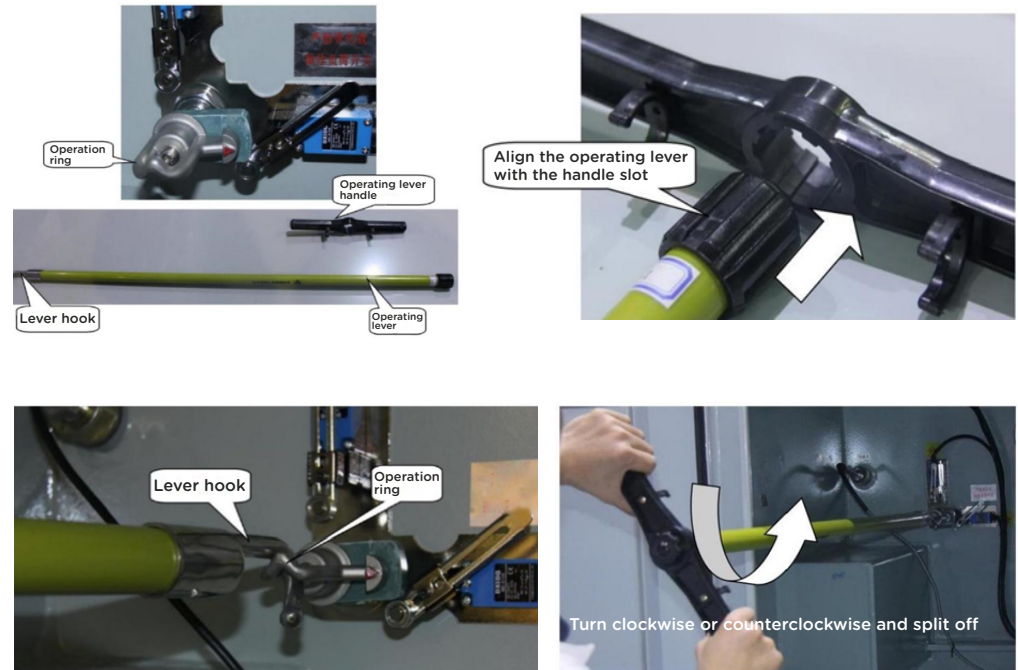


3.2.3 Load Switch

- 1) This switch cannot be used to open the fault current.
- 2) This switch can be turned clockwise or counterclockwise.
- 3) First close the special removable operating lever handle with the operating lever, then hook the load switch operating loop, then press the rotating mechanism, and turn the operating lever until the load switch is fully positioned in the split or close position.

It is strictly prohibited to operate the load switch with load, which is only operated under no- load condition. If the load switch is stuck at a certain position, please stop operating the switch and consult the manufacturer of the box transformer.

4) The corresponding photos are detailed as follows:



3.2.4 Gas relay bleeding



- ① Open the cover of gas relay and remove the plug of gas drain plug



- ② Unscrew the gas relay vent valve for ventilation, the method is: slowly loosen the vent valve bolt, but do not screw, the tightness is appropriate, do not touch the probe of the red cap.
- ③ When the oil is released in the bleed plug, it means that there is no gas in the gas relay. At this time, tighten the vent valve, then spin on the bleed plug, and clean the oil stains with a cloth, and tighten the outer cover bolt.

3.2.5 Transformer Oil Release- Replenishment

If the transformer oil level indication is above or below the warning oil level line, the transformer shall be drained or replenished.

Steps of oil discharge:

- 1、 Before oil release, take oil samples for test, measure the insulation strength and water content of the transformer oil, and confirm whether the transformer is damp;
- 2、 Transformer oil shall be placed in a sealed and clean special oil tank, and transformer oil shall be injected into the oil tank from the bottom valve of the oil tank;
- 3、 Measure the relative humidity of the air during oil discharge, and record the start time of oil discharge;
- 4、 The exposed area of the transformer body shall be reduced when releasing the oil.

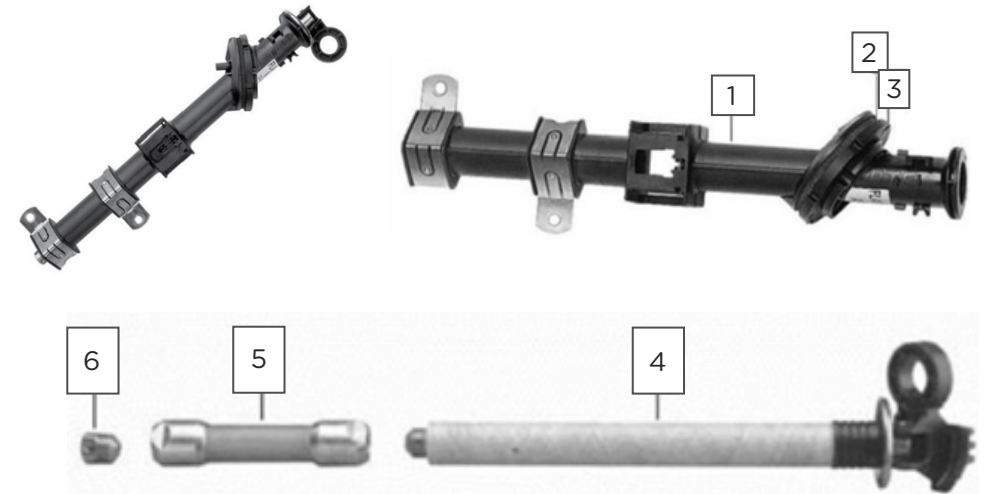
Step of oil filling:

- 1、 Prevent oil mixing, the added oil should pass the test;
- 2、 Before filling, the heavy gas protection device shall be changed to the signal level to prevent the switch (if have);
- 3、 Check the gas relay, replenish the gas and release it in time. If there is no abnormal phenomenon after 24 hours, connect the heavy gas to the switch position (if have);

- 4、 The oil replenishment amount shall not be excessive or insufficient, and the oil level should be adapted to the oil temperature of the transformer at that time;
- 5、 It is forbidden to replenish oil from the valve at the lower part of the transformer to avoid entrainment of bubbles;
- 6、 The time interval between oil replenishment and power transmission of the transformer shall be more than 12 hours to ensure the escape of bubbles in the oil.

3.3 Fuse Installation

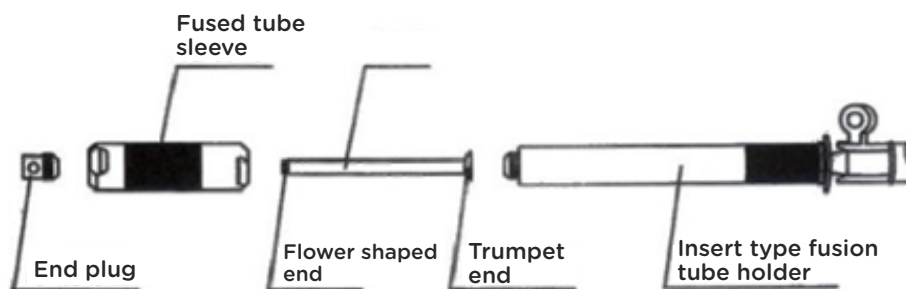
3.3.1 Small current protection fuse is inserted into Bay-O-Net installed obliquely on the side of American box transformer with voltage level of 23 kV / 40.5 kV, as shown in the following figure:



No.	Name
1	Melting pipe bracket
2	Box wall sealing pad
3	Install nuts
4	Plug in fusion socket
5	Fusion sleeve
6	End screw plug

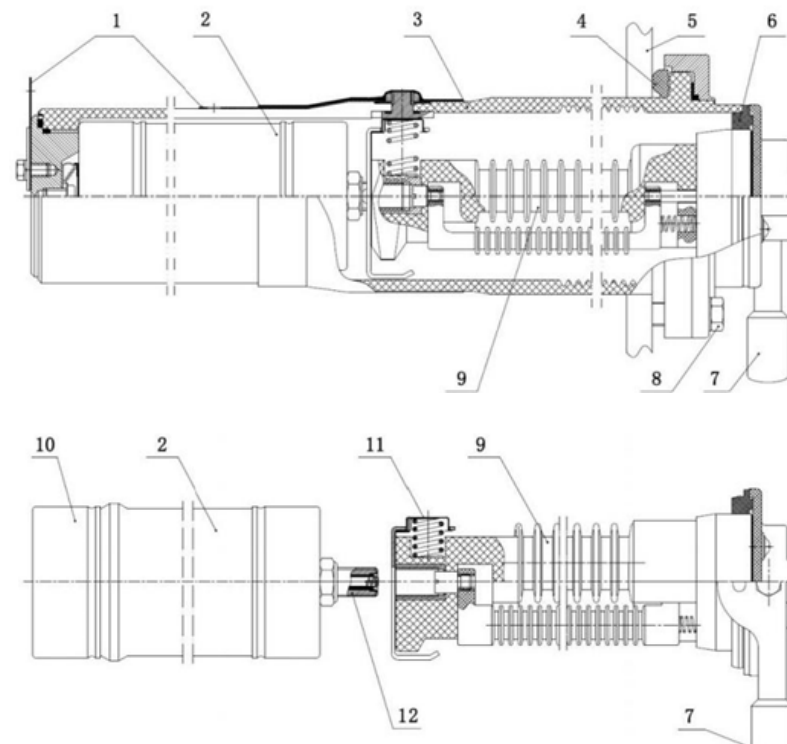
- 1- When removing items 4,5 and 6 from the assembly, the transformer must be powered off and release pressure;
- 2- The melt pipe can be conveniently replaced at the site in the following steps (the internal disassembly components are shown below):

- 1) Pull the entire insert fuse seat from the fuse bracket and wipe the oil trace clean.
- 2) Unscrew the fuse sleeve and end plug from the fuse seat with a wrench.
- 3) Remove the fuse from the tube sleeve (if the outer insulation "Teflon" is not melted, the floral end must be straightened to remove the melt from the tube sleeve).
- 4) Check the inside of the fuse sleeve to make sure it is clean, and then load the new fuse tube.
- 5) Screw the fusion tube sleeve installed with the fusion tube onto the fusion tube base.
- 6) Turn the end plug on the fuse sleeve, and then unscrew the end plug to confirm that the flower end of the melt is evenly opened, and finally tighten the end plug on the melt sleeve.
- 7) Insert and install the replaced plug-in fusion tube into the fusion tube support.



3.3.2 The full range protection current limiting fuse shall be horizontally installed and inserted in the domestic American box transformer with voltage level of 40.5 kV:

- 1- Composed of fuse, fuse base and load melt (handle and contact), plug-in structure, fuse can be easily replaced on site. Overall structure and internal installation diagram are shown below:



No.	Name
1	Connecting terminal
2	Fuse
3	Support shell components (base)
4	Seals (fuel tank side)
5	Install the panel
6	Sealing ring (fuse side)
7	Handle
8	Fixing bolt
9	Melt carrying part (pull rod)
10	Rear conductive cap
11	Front conductive cap
12	Fuse and melt bolts

2、Specific steps to replace the fuse parts are as follows:

Special warning:

In order to prevent overvoltage and wrong operation caused by fuse explosion, please note that when the fuse needs to be replaced, it is necessary to detect and troubleshoot. After confirming that there is no fault on the line, the new fuse can be replaced only if the fuse is not charged. After replacement, the following matters should be paid attention to when switching the switch:

- 1) Operation and auxiliary personnel shall not be facing the outer cylinder of the fuse;
- 2) Leave a sufficient safe distance and operate from the side with the operating lever;
- 3) Wear safety protective gear (safety helmet, safety mirror, etc.).

In order to prevent the power transmission, due to the failure caused by the fuse hit out.

Please strictly abide by the above items and paste the above content beside the transformer switch, please inform the wind farm users in time.

- 1) Wear clean cotton gloves when replacing the fuse (to prevent the pull rod or fuse from being polluted during operation and affecting the insulation); Disconnect the signal output wiring before pulling out the fuse; There is no need to loosen 4 fixing bolts;
- 2) Then remove the fuse and the pull rod from the base. When extracting the fuse, lift the handle first and slowly remove the fuse from the base. Inside out, the other hand in the fuse out of about 200mm to hold, and then all out;
- 3) Use M16 open-ended wrench or directly screw the fuse, loosen the fuse and pull rod, and replace with a new fuse;
- 4) Before inserting the fuse, first check whether the threaded connection between the fuse and the carrier melt is loose, and the end face of the metal outer cap of the fuse and the surface of the conductive contact on the carrier melt are clean. Then make the arrow direction of the carrier melt cover consistent with the arrow direction of the aluminum gland, hold the handle with one hand and hold the fuse with the other hand, slowly insert it into the fuse support, pull down the handle and lock it after it is inserted in place;
- 5) Secure the signal output terminal in the tank panel;
- 6) In the three-phase system, when the one-phase fuse action, the three-phase fuse should be replaced.

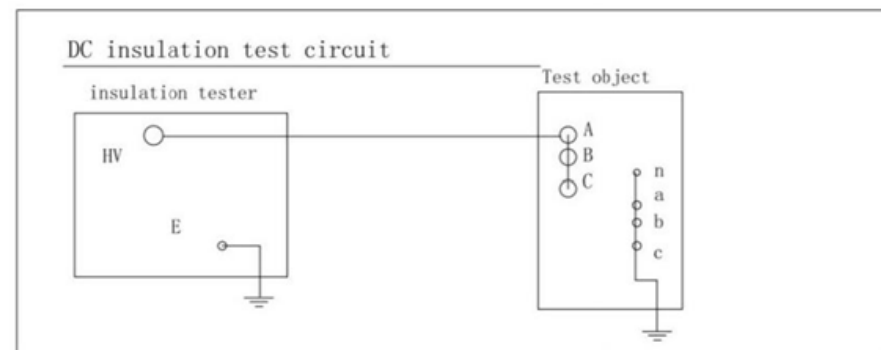
4.Test Before Operation

4.1 Insulation Measurement

- The insulation resistance meter with 2500V and measuring range not less than 10000m Ω shall be used for measurement, and the accuracy of the insulation resistance meter shall not be lower than level 1.5.
- Press the test site on the table for testing:

Serial No.	Two-winding transformer		Low-voltage double-split transformer	
	Tested winding	Grounding position	Tested winding	Grounding position
1	High voltage	Low voltage and oil tank	High voltage	Low voltage 1、 Low voltage 2 and oil tank
2	Low voltage	High voltage and oil tank	Low voltage 2	High voltage、 Low voltage 2 and oil tank
3	High voltage and low voltage	Oil tank	High voltage and low voltage 2	High voltage、 Low voltage 1 and oil tank
4	1	/	Low voltage 1 and Low voltage 2	High voltage and oil tank

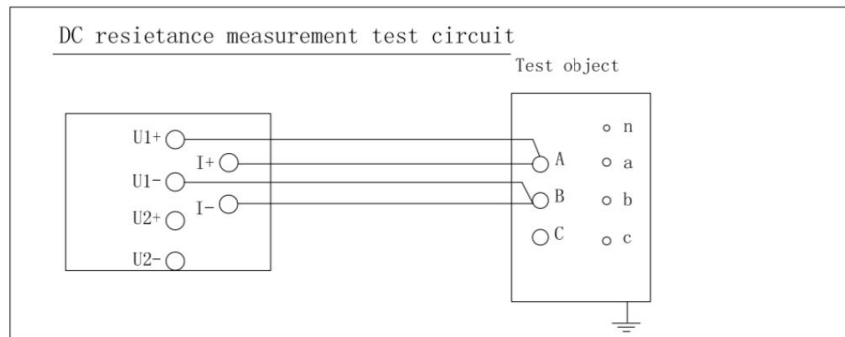
- Wiring diagram:



4.2 Winding resistance measurement

- The resistance of each winding of the transformer shall be measured at its line end, and the test current shall generally not exceed 10% of the rated current;
- When the three-phase transformer winding is connected without neutral point, the line resistance should be measured, such as AB, BC and CA. If there is a neutral point, the phase resistance should be measured, such as AO, BO and CO;
- When the winding is D-connected, measure the wire resistance;
- Winding temperature must be accurately recorded during winding resistance measurement. For oil-immersed transformers, the average oil temperature can be taken as the winding temperature when the oil temperature is stable.

- Wiring diagram:



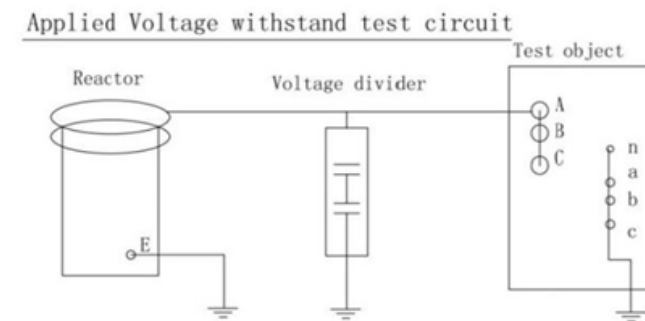
4.3 Exterior Pressure Resistance Test

- Before external pressure resistance, the tested fuel tank shall be reliably grounded and confirm that the oil level is normal;
- The frequency of the foreign affairs withstand voltage test shall not be less than 80% of the rated frequency, the voltage waveform shall be close to the sine wave, and the ratio of peak and RMSM shall not be greater than 2 ± 0.07 ;

- Press the test site on the table for testing:

Serial No.	Two-winding transformer		Low-voltage double-split transformer	
	Tested winding	Grounding position	Tested winding	Grounding position
1	High voltage	Low voltage and oil tank	High voltage	Low voltage1、 Low voltage 2 and oil tank
2	Low voltage	High voltage and oil tank	Low voltage 1	High voltage、 Low voltage 2 and oil tank
3	1	1	Low voltage 2	High voltage、 Low voltage 1 and oil tank

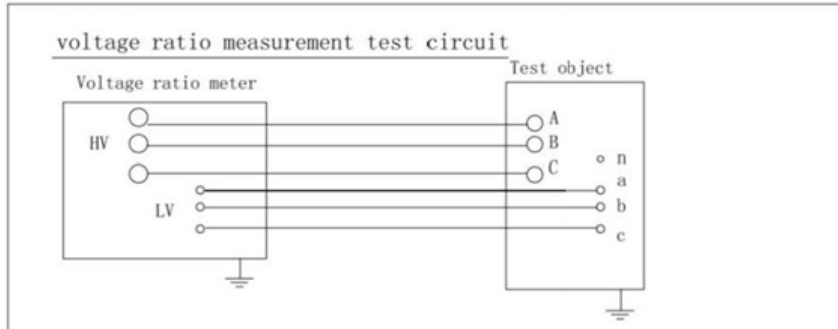
- Wiring diagram:



4.4 Voltage Ratio Measurement

- The voltage ratio measurement is recommended to use a voltage ratio bridge with an accuracy of no less than the 0.1 level;
- For double winding transformer, test the voltage ratio and connection group of high voltage to low voltage;
- For low voltage double split transformer, measure the voltage ratio and connection group of high voltage to low voltage 1 and high voltage to low voltage 2 respectively;
- The measurement shall be performed separately on each tap.

- Wiring diagram:

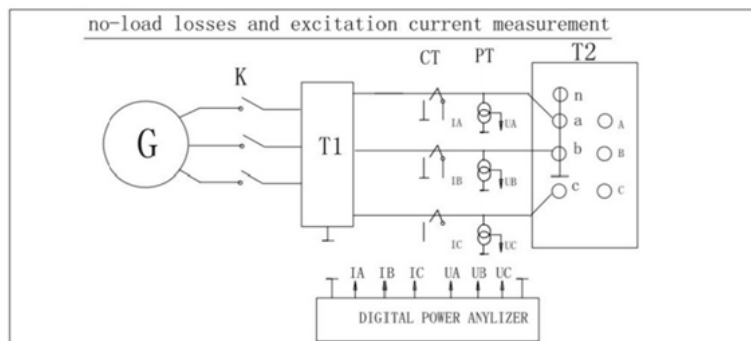


4.5 No-load loss and no-load current measurement

- The measurement of no-load loss and no-load current shall be supplied to the rated voltage (as symmetrical as possible) from the line end of each winding (generally low-voltage winding (as far as possible), and the remaining windings are open;
- The high-voltage winding is placed in the main tap;
- The fuel tank is reliably grounded;
- The accuracy of the transformer used in the no-load test shall not be less than 0.2 level, and the instrument accuracy used shall not be less than 0.5 level;

When the voltage waveform is distorted, the mean voltage U' and RMS voltage U are inconsistent, the applied voltage shall be subject to the mean voltage, and the deviation shall not exceed 3%, No-load loss shall be corrected, $P_o = P_m \cdot (1 + (U' - U/U'))$

- Wiring diagram:



4.6 The rest of the test items

- Test item 1 and test method (example: AC LV side voltage, measured using a multimeter at around 500Vac)
- Test item 2 and test method (example: AC LV side voltage, measured using a multimeter, ensuring at around 500Vac)

5. Power on and put into operation & Power down maintenance

5.1 Power on and put into operation

5.1.1 Inspection before power on and operation

Step 1: Check whether the nameplate of the transformer complies with the actual requirements of the transformer. For example, whether the side voltage level, connection group, capacity and operation mode are consistent with the actual requirements.

Step 2: Check the transformer body and all accessories and all parts of the fuel tank for defects.

Step 3: Check whether the outlet casing and bus on both sides of the transformer high and low voltage are well connected, whether the color mark of the three phases is correct, whether the transformer grounding is firm, and whether the tap switch position is correct.

Step 4: check whether there is gas inside the gas relay. If there is any gas, discharge it completely and check whether the contact can act accurately.

5.1.2 Power on and operation steps

Step 1: Ensure sufficient operating distance, hook the operating hole with the operating rod, align the conductive rod with the casing guide hole, and push forward, until you feel the miso between the elbow head and the casing, the elbow head and the casing are closely combined.

Step 2: Operate the load switch with the insulation operating rod, and do not start the transformer with the load.

Step 3: After the transformer is charged, strictly operate the tap switch.

Step 4: After power transmission, the low-voltage side uses the multimeter to check the electricity and check whether the three-phase voltage is symmetrical.

Step 5: Check whether the transformer noise is normal and whether the oil thermometer indicator is normal.

5.2 Power down maintenance

Lower power maintenance step:

Step 1: Disconnect the low-voltage circuit breaker switch first, and observe whether the three- phase ammeter indication is zero.

Step 2: Operate the high-voltage load switch to power off the transformer. Load switch cannot be operated with transformer fault and low voltage short circuit.

6. Routine Maintenance

6.1 Eeneral maintenance

(Give items that can be maintained without a power outage)

List of projects	Inspection method	Inspection cycle
Operation status	Current voltage load ambient temperature	Once a day
Transformer temperature	Record the value displayed by the thermometer	Once a day
Abnormal noise or vibration	Whether there is power generation sound caused by poor grounding, and whether there is abnormal sound and resonance in accessories	Once a day

6.2 Parts Maintenance

List of projects	Power off steps	Inspection method	Inspection cycle
Pressure valve	See 5.2	Check for any oil seepage or oil injection, and whether the marking rod is moving	Once a month
Gas relay (If have)	See 5.2	Check whether there is oil seepage, whether there is gas accumulation, if the gas accumulation needs exhaust treatment	Once a month
High and low voltage bushings	See 5.2	Check for any oil seepage and cracking	Once a month
Non excitation tap changer	See 5.2	Check for oil seepage	Once a month
Radiator, fuel tank flange	See 5.2	Check for any oil seepage and cracking	Once a month
Respirator (If have)	See 5.2	Check whether the silica gel has discoloration, and 2/3 of the discoloration needs to be replaced; Check whether the breathing is unblocked; The oil level in the oil cup is between the maximum and minimum	Once a month
Fastening bolt	See 5.2	Check whether the bolts are loose. If they are loose, tighten them	Once a month

6.3 Component replacement

6.3.1 Replace the heat exchange unit (not applicable)

(Provide the replacement steps and the drawings)

6.3.2 Replace the measurement and control device (not applicable)

(Provide the replacement steps and the drawings)

6.3.3 Replace the temperature and humidity controller (not applicable)

(Provide the replacement steps and the drawings)

6.3.4 (Other matters not listed)

(Provide other component replacement steps and drawings)

6.3.5 Replace oil level temperature controller for transformer.

Replace steps:

- 1) Open the dial counter-clockwise, remove the secondary line in the dial, remove the gram head, and properly keep the above items;
- 2) Cut the binding band of the capillary;
- 3) Remove the three mounting bolts from the dial and remove the temperature control gauge;
- 4) Remove the three mounting bolts of the dial and remove the temperature control meter;
- 5) Take a new oil level temperature controller, insert the temperature bag into the thermometer base, and check the oil level in the thermometer base (do not screw the thread, and the oil immersion length of the temperature bag after pulling out is about 10mm). After the oil level is qualified, tighten the mounting bolt first, and then the small bolt on the upper part of the temperature bag;
- 6) Fix the temperature control dial with three bolts;
- 7) Tie the capillary;
- 8) Open the dial counterclockwise, remove the transport protective pad (with eye sign) in the dial, and install the gram head;
- 9) Set the temperature controller according to the requirements of the drawing and test the secondary line after passing the test;
- 10) Adjust the red pointer (historical maximum temperature) to the right side of the grey pointer;
- 11) Cover the dial and tighten clockwise.



6.3.6 Replace yzf3 oil level gauge ...

Replace steps:

- 1) Open the screw of oil level gauge junction box, open the junction box, remove the secondary wire in the junction box and remove the gland;
- 2) Drain the oil from the transformer until the scale pointer of the oil level gauge is 0;
- 3) Remove the four mounting bolts of the oil level gauge and take out the oil level gauge;
- 4) Take a new oil level gauge, assemble the floating ball, put it into the box in place, and fasten the mounting bolts;
- 5) Unscrew the oil level gauge junction box screws, open the junction box, connect the secondary wire in the junction box, and tighten the junction box screws after installing the gram head;
- 6) Fill the transformer with oil to the oil level corresponding to the oil temperature;
- 7) Open the exhaust plug on the upper oil pillow and remove the moisture absorber;
- 8) Inflate the capsule from the interface of the moisture absorber. After the gas in the oil is discharged and the oil seeps from the exhaust plug on the upper part of the oil conservator, tighten it and wipe the oil stains dry;
- 9) Reassemble the moisture absorber;
- 10) Check again whether the oil level corresponds to the oil temperature.



7.Troubleshooting

According to:

Level 1 is critical, which may directly correspond to the shutdown fault of transformer

Level 2 is important, which may correspond to the transformer shutdown failure, but it generally does not happen immediately

Level 3 is the general, the transformer operation defects, but generally will not cause the transformer shutdown failure

Level 4 is minor, and the normal operation of the transformer is generally not affected by the general operation defects of the transformer

Fault name	Fault grade	Possible causes	Investigation method
Pressure relief fault	Level 1	Secondary line fault	Transformers in appearance without oil injection Secondary line fault and insulation defect
		Respiratory blockage (not applicable)	The respirator valve and cargo main pipe valve are closed
		Oil level is too high	The oil level is highly abnormal, resulting in an internal overpressure
		Internal fault	When the transformer is injected with oil, the oil sample is found to be black; Abnormal found in electrical test: such as obvious change in winding resistance, abnormal variable ratio, low voltage no-load abnormality (feedback diagnosis) The transformer shell has bulging deformation, cracking and other defects
Heavy gas trip	Level 1	Secondary line fault	Heavy gas action knowledge indicates no action Secondary line fault fault and insulation defect
		Internal fault	Heavy gas action indication has been activated Abnormalities found in electrical test: such as obvious change of winding resistance, abnormal transformation ratio, low voltage and no-load abnormality (feedback diagnosis)

Fault name	Fault grade	Possible causes	Investigation method
Light gas alarm (not applicable to air cushion structure)	Level 3	False alarm	Gas gas is below the alarm limit Failure and insulation defects are found in the secondary line
		Internal residual gas leakage	
		Internal faults or electrical defects	
Oil temperature super high trip, high oil temperature alarm	Level 2	Temperature control fault	Measure the actual oil temperature without exception Show oil temperature obvious discord logic
		Disability of heat dissipation device	Infrared measurement of the heat dissipation device temperature is in an abnormal operating condition
		Overload operation	The abnormal oil temperature is true; Transformer excess fixed capacity operation
		Internal and winding fault	The abnormal oil temperature is true; The transformer load is normal, with no obvious capacity over the limit Abnormal low voltage no-load current and loss of transformer (feedback diagnosis)
High oil level	Level 3	Oil level calculation reason	Infrared measurements showed normal oil levels and the oil level gauge was abnormal
		Actual refueling is too high/insufficient compensation rate	Oil temperature is normal and infrared measurement shows that oil level is indeed high (feedback diagnosis)
		Oil temperature is abnormal	Infrared measurement shows that the oil level is indeed high, and it is found that the transformer oil temperature is also significantly high

Fault name	Fault grade	Possible causes	Investigation method
Low oil leve	Level 1	Oil level calculation reason	Infrared measurements showed normal oil levels and the oil level gauge was abnormal
		Low refueling level/ insufficient compensation rate	No oil leakage was found Oil temperature is normal and infrared measurement is indeed low (feedback diagnosis)
		Extreme low temperature weather	No oil leakage was found In case of extreme low temperature weather, no-load or light load of transformer is common
		Leakage oil	Significant oil leakage was found Infrared measurement shows that the low oil level is true
Heat exchange device failure (not the content of bare metal transformer)	Level 2	Power failure	Out of service, no display
		Circuit failure	The panel shows the fault category and controls the instruction manual
Low voltage incoming circuit breaker trip (not bare content of transformer)			
Temperature and humidity controller over- temperature? Fault (not part of transformer)			
Circuit breaker failure? (Not the bare machine content of the transformer)			

Fault name	Fault grade	Possible causes	Investigation method
Load switch fault? (Combined with the actual switch form and configuration writing. Design responsibility)		Oil seepage at the flange core position	Tighten the gasket at the flange core
		The operating mechanism fails	Manually switch the load switch, the feel is very light, and the switch switching sound can not be heard.
Tap changer fault	Level 1	The switch shaft cannot rotate	Try to turn the switch and find that the torque is too large to turn to the required gear
		The gear switch does not reach the specified position	Check found an abnormal position of the switch operating handle Electrical tests such as transformer winding DC resistance and winding transformer ratio may be abnormal
		Switch scattered	Try to turn the switch, found that the torque is very small, there is no obvious gear feel The winding resistance and transformation ratio are obviously abnormal when changing the gear of the switch (the gear change data remains unchanged)
The thermometer reading is incorrect	Level 3	The temperature meter fails	The obvious fault is inconsistent with the reading No installation and wiring
		Wrong thermometer installation or wiring	Control the instructions or comparison, and find obvious differences or errors
		Temperature-controlled leakage using a capillary structure	The temperature display does not vary with the temperature change Significant capillary damage was found
		Line fault with PT100 temperature control	Found wiring errors, or damage
		Mechanical damage	Check the temperature control system and find obvious signs of external damage

Fault name	Fault grade	Possible causes	Investigation method
Transformer oil leakage	Level 2	Residual oil mark due to installation or other reasons	The insulation solution is dark and has a bright color difference with the normal short-term leakage solution (For example, the transformer oil color is obviously yellow) No extended traces were observed for an extended time
		Welding leakage	The problem area corresponds to the weld, and there is insulating liquid inside the weld Check for visible and obvious welding defects
		Sealing leakage	Leakage or expansion The problem area is sealed with insulating solution inside the seal Abnormal seal state is found: abnormal installation state, aging, damage, etc
Voltage imbalance	Level 2	Measurement system problems	Alignment measurement readings were normal
		Right to ground impedance imbalance leads to ground voltage imbalance	Compare the voltage value of the measuring line Observe the coincidence rate, the load three-phase imbalance
		Intake line fault, or transformer winding fault	Measure phase voltage and line voltage to determine whether there is phase absence (feedback diagnosis)
Abnormal noise during operation	Level 3	Harmonic current influence	Observe the abnormal sound changes at different load rates
		DC partial magnetic, or even harmonic influence	Noise was significantly abnormal When the load change and the time change, the noise changes significantly
		Loose the external parts of the transformer	Close to the observation, you can find that a position of the sound is obviously loud The sound is very similar to that caused by external trunking and loose components
		Internal loosening, or iron core displacement	Abnormal noise without obvious change for a long time No obvious change in the direction of use

Fault name	Fault grade	Possible causes	Investigation method
Fuse blown	Level 1	Short circuit large current	Check the relevant operation information, there is a high current short circuit High and low voltage switches have action information
		Operating current with high overheating	Check the relevant operation information, there is a large running current
		Wrong fuse specification	Fuse specifications are not consistent with the requirements
		Fuse installation is loose	Obvious heating marks at the contact installation position Check if the card installation status is abnormal
A short circuit to ground	Level 1	Insulation faults or defects	Abnormal voltage of different phases Check the insulation resistance after shutdown, which may be abnormal
		External abnormal short circuit	Abnormal voltage of different phases External short wiring not removed, or abnormal ground flashover
		Overvoltage protection device fault grounding	Abnormal voltage of different phases Check the insulation resistance may be abnormal after shutdown. Check the insulation resistance may be abnormal after shutdown The overvoltage protection device of infrared patrol inspection may have abnormal high temperature, or damage is found through visual inspection

8.Appendix

Tightening torque table

1、Reference table of common national standard bolt torque

Bolt	Torque(N.m)	Screw	Torque(N.m)
M8	20	M22	370
M10	30	M24	480
M12	50	M27	720
M14	95	M30	860
M16	130	M33	1030
M18	195	M36	1230
M20	240	M39	

2、Reference table for torque of bolts commonly made in USA

Bolt(USA)	Torque(N.m)	Screw	Torque(N.m)
1/4	8	3/4	200
5/16	20	7/8	370
3/8	30	1	480
7/16	40		
1/2	50		
9/16	95		
5/8	120		

Note: 1. The strength grade of the above bolts is grade 8.8;
2. The connector is made of non steel material, and the assembly torque value needs to refer to the instructions in the annex.



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